Matroid Intersection Theorem

Thm: $M_1 = (E, \mathcal{I}_1)$, $M_2 = (E, \mathcal{I}_2)$ matroids

then the max size of a set in $\mathcal{I}_1 \cap \mathcal{I}_2$

is: $$z^* = \min_{U \subseteq E} : r_1(U) + r_2(E \setminus U)$$

Pf: "$\leq$" is immediate:

$\forall I \in \mathcal{I}_1 \cap \mathcal{I}_2$, $\forall U \subseteq E$

$$|I| = |I \cap U| + |I \setminus U| \leq r_1(U) + r_2(E \setminus U)$$

Proof — continued —

We prove reverse inequality by induction

on $|\{e \in E : r_1(e) = r_2(e) = 1\}|$

Exercise: Check the base case: card $= 0$.

Pick some e with $r_i(e) = 1$, $i = 1, 2$.

Consider: $M_1 \setminus e$, $M_2 \setminus e$. The inductive hypothesis applies.

Note: for deletion, the rank fns are unchanged.

If $\min_{U \subseteq E \setminus e} : r_1(U) + r_2((E-e) \setminus U) \geq z^*$

By ind. hyp: $\exists I \subseteq E - e$, $I \in \mathcal{I}_{1 \setminus e} \cap \mathcal{I}_{2 \setminus e} \Rightarrow I \in \mathcal{I}_1 \cap \mathcal{I}_2$.

Thus: $|I| \geq z^*$. So we must have $\hat{U} \subseteq E - e$

with: $\boxed{r_1(\hat{U}) + r_2((E-e) \setminus \hat{U}) \leq z^* - 1}$

Proof — continued —

Next, consider: M_1/e, M_2/e.

For same e.

Recall: $I \in \mathcal{I}_{1/e} \iff I+e \in \mathcal{I}_1$, similarly for $\mathcal{I}_2$.

$$r_{M_1/e}(U) = r_{M_1}(U+e) - r_{M_1}(e).$$

Ind. Hyp applies to both M_1/e, M_2/e: hence, $\exists\, I \in \mathcal{I}_{1/e} \cap \mathcal{I}_{2/e}$

$$|I| = \min_{U \subseteq E-e} \underline{r_{M_1/e}(U) + r_{M_2/e}((E-e)\setminus U)}$$

If $\boxed{RHS} \geq z^* - 1$ then $I \in \mathcal{I}_{i/e} \Rightarrow I+e \in \mathcal{I}_i$ $i=1,2$

$\Rightarrow |I+e| \geq z^*$.

Hence let ~~us~~ us consider the remaining case: $RHS \leq z^* - 2$

U

Proof — continued —

Hence:

$$\min_{U \subseteq E-e} : r_{M_1/e}(U) + r_{M_2/e}\big((E-e)\setminus U\big) \le z^* - 2$$

$$\Rightarrow \exists\, \tilde{U} \subseteq E-e :$$

$$\underline{r_{M_1/e}(\tilde{U})} + r_{M_2/e}\big((E-e)\setminus \tilde{U}\big) \le z^* - 2$$

$$\Rightarrow \quad r_{M_1}(\tilde{U}) - 1 + r_{M_2}(E \setminus \tilde{U}) - 1 \le z^* - 2$$

$$\Rightarrow \quad \boxed{r_{M_1}(\tilde{U}+e) + r_{M_2}(E\setminus \tilde{U}) \le z^*}$$

Recall from deletion: $\boxed{r_{M_1}(\hat{U}) + r_2\big((E-e)\setminus \hat{U}\big) \le z^* - 1}$

Proof — continued —

Adding the inequalities:

$$2z^* - 1 \geq r_1(\hat{U}) + r_1(\tilde{U}+e) + \left(r_2((E-e)\setminus\hat{U}) + r_2(E\setminus\tilde{U})\right)$$

$$\geq \underline{r_1(\hat{U}\cup\tilde{U}+e) + r_1(\hat{U}\cap\tilde{U}) + \left(r_2(E\setminus\hat{U}\cap\tilde{U}) + r_2(E\setminus(\hat{U}\cup\tilde{U}+e))\right)}$$

follows from: $e \notin \hat{U}\cup\tilde{U}$

and submodularity of rank fns.

But: $z^* \leq r_1(\hat{U}\cap\tilde{U}) + r_2(E\setminus(\hat{U}\cap\tilde{U}))$

$z^* \leq r_1(\hat{U}\cup\tilde{U}+e) + r_2(E\setminus(\hat{U}\cup\tilde{U}+e))$

$\divideontimes$ $\blacksquare$

Conclusion

This proof is due to Woodall.

Can also prove using LP duality.

Bipartite Matching is a matroid intersection.

We can use the theorem to redrive

Hall's Marriage Thm.